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Short-Listed Filling Materials for Transparent PMMA

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Introduction

Propery of Polymethyl methacrylate(PMMA) Object

PMMA is solvent-sensitive and susceptible to surface damage such as scratches, chipping, and gouging.

Three Research Questions

What is a suitable adhesive to fill scratches for transparent PMMA?

- Refractive Index(RI): similar to PMMA (RI 1.492)
- Adhesive strength: moderate strength required
- Viscosity/concentration: low viscosity, less molecular weight preferred
- Shrink/Evaporation of solvent: minimum shrink/evaporation
- Aging behavior: staying transparent / no yellowing
- Reversibility

What is a proper application method?

- Capillary action or brushing

Does the crack size matter?

Case Study [1]

A bracelet is an object from the study collection of the Cultural Heritage Agency of the Netherlands (RCE), that is naturally aged. The date is unknown, it shows stress cracks and fracture. This poster focuses on the filling cracks.

Fig 1. Stress cracked bracelet[2]

Damage description

- Multiple cracks
  - Vertical direction
  - Diagonal direction
- Cracks possibly caused by
  - High relative humidity,
  - Solvent exposure,
  - Release of internal stress

Purpose of the treatment

- Making cracks Invisible
- Keeping transparent color with smooth surface
- Improving the structural strength

Research

Selecting suitable adhesives by properties[4]

Adhesive		RI	Reversi-bility	Molecular weight	Viscosity of product
Epoxy Resin	Hxtal-NYL-1	1.516	N	Low (resin+catalyst mixture)	200-300 cP.
Hydrocar-bon resin	Regalrez® 1094	1.518	Y	928 g/mol[6]	Low (40% w/w in ShellSol D40)
Hydrocar-bon resin + sebacate	Regalrez® 1094 + Tinuvin® 292	1.518	Y	508.78g/mol[7]	Low (40% w/w Regalrez in ShellSol D40 with 2% resin Tinuvin)

- The ability of a resin to level well is directly related to the size and structure of the polymer chain length or molecule - during solvent evaporation, the low molecular weight resin can level with the surface better.[8]
- Selected adhesives exhibited good light aging properties.
- The addition of Tinuvin improved the stability of resins for aging.
- ShellSol D40 is a solvent, but the lack of polarity in ShellSol D40 makes it unsuitable for breaking down the molecular structure of PMMA. So, it is safe to use with adhesive for filling PMMA.

Reasons of non-selection

• Acrylic resin adhesives

It was tested with 20% solution in low-evaporating solvents. Because of the high viscosity of acrylics, it was difficult to have capillary action (e.g. molecular weight of Paraloid B67 is 60,000 g/mol, viscosity is 1000-2500 cP.[9]). Lowering the concentration will expose PMMA to more solvents, which is not ideal.

• UV curing adhesives

UV curing is not always possible due to location or shape constraints. It also had issues during tests; atmospheric oxygen inhabiting the cure on the surface layer, shrinking during polymerization, and yellowing.

Tools for filling cracks and areas of loss

- Mini brush 3050R size 20/0[10]
- Insect needles: depending on the viscosity of materials for capillary action

Result

Fig 2. Bracelet after the treatment[3]

Adhesive used

Hxtal-NYL-1

Application Method

Capillary action: flowing into cracks

Visual accessment

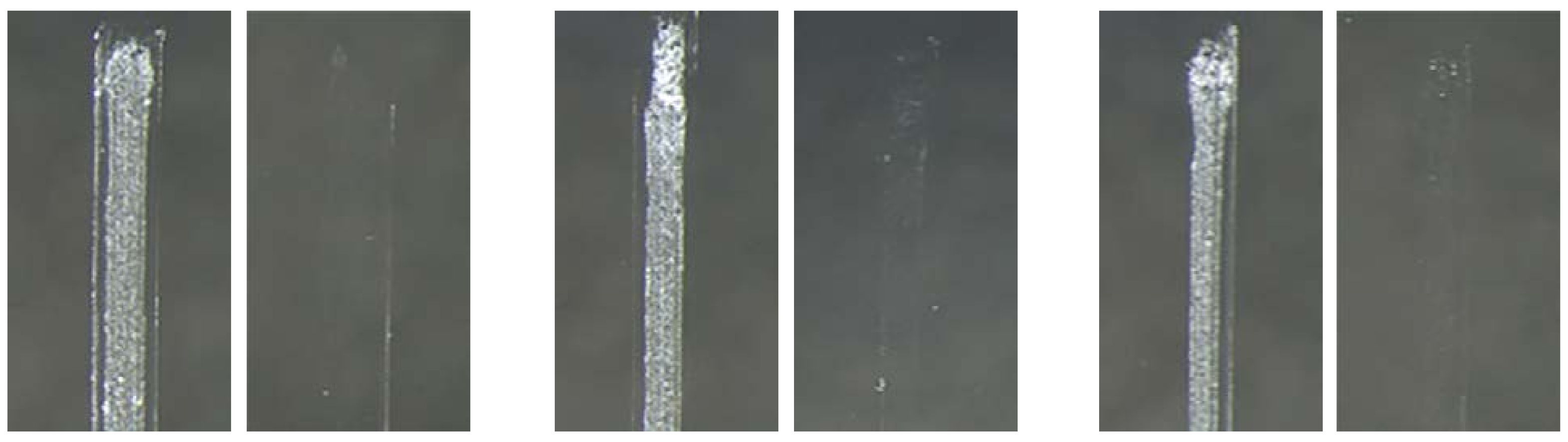
Only visible as hairlines

Remarks

RI difference between PMMA and Hxtal-NYL-1 is 0.024, but for small cracks after treatment looked good. When bigger chips were filled with the adhesive, it showed the RI difference.

Adhesives with good results in tests[5]

•Microscope image x25, before& after treatment



Hxtal-NYL-1 Regalrez® 1094 Regalrez® 1094 + Tinuvin® 292

Adhesive	Advantage	Disadvantage
Hxtal-NYL-1	• Low viscosity • Shrinking less than 1% • Color difference (ΔE94) after aging is very low: 0.79	• Not reversible • Bubbles
Regalrez® 1094	• Low viscosity • Reversible	• ΔE94 after aging is a bit higher than Hxtal-NYL-1: 1.8

Conclusion & Future Research

The case study evaluated various adhesives for transparent PMMA objects, focusing on their refractive index, viscosity, and aging properties. Hxtal-NYL-1 and Regalrez® 1094, with and without Tinuvin® 292, were tested for their suitability for filling cracks and scratches. Hxtal-NYL-1 has low viscosity and minimal shrinkage, with an excellent refractive index match for small stretches. However, its lack of reversibility and potential for bubbles present challenges. Regalrez® 1094 and Regalrez® 1094 with Tinuvin® 292 have good viscosity and are reversible. They showed slightly higher color change upon aging but remained acceptable for use. The low viscosity of adhesives allows them to flow into the cracks with good leveling; using capillary action is recommended. Furthermore, future research can explore optimizing application techniques because using capillary action makes it hard to control adhesives on the surface.

Reference: **1,2,3** Anna Laganá, Thea B. van Oosten. *Back to transparency, back to life: research into the restoration of broken transparent unsaturated polyester and poly(methyl methacrylate) worKs of art.* ICOM-CC 16th: Triennial Conference in Lisbon "Cultural Heritage/Cultural Identity: The Role of Conservation" 2011. **4,5** Anna Laganá , Julia Langenbacher, Rachel Rivenc, Maelle Car o. *The future of looking younger: A new face for PMMA. Research into fill materials to repair poly(methyl methacrylate) in contemporary objects and photographs.* ICOM-CC 18th Triennial Conferen ce Copenhagen. "Linking Past and Future " 2017. **6** https://www.synthomer.com/Media/tds/Regalrez%201094%20Hydrocarbon%20Resin.pdf **7** https://dispersions-resins-products.basf.us/files/techn ical-datasheets/Tinuvin_292_October_2019_R4_IC_PP.pdf **8** Julia M. van den Burg, Kate Seymour, "*Varnishing and Inpainting/Retouching*", Cultural Heritage Agency of the Netherlands, 2022. p.15 **9** https://www.palmerholland.com/getmedia/278a7c7b-ef2c-4491-b92e-84bc88f3af12/MITM04031_2 **10** https://www.amazon.com/Princeton-Synthetic-Miniature-Watercolor-3050R-12/dp/B0009ILH5 U?th=1

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